

Aditya Hansoty

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EDUCATION

University of Michigan

B.S. Biomedical Engineering, Minor in Computer Science | GPA: 3.73/4.0

Activities: Biomedical Engineering Society (Executive Board)

Ann Arbor, MI

Aug. 2025 – May 2029

EXPERIENCE

Systems Engineering Intern

Naval Medical Research Unit San Antonio, Department of Navy

Incoming Fall 2026

San Antonio, TX

Prosthetics Research Intern

Rehabilitation Robotics Lab, University of Illinois Chicago

May 2026 – Sept. 2026

Chicago, IL

- Designed and iterated threaded encoder caps for a powered ankle-foot prosthesis in SolidWorks, refining geometry and tolerances across 3D-printed and CNC-machined revisions
- Controlled a powered ankle-foot prosthesis via TwinCAT and led Vicon motion capture across 10 human-subject trials; built Python pipelines synchronizing EMG, ECG, and metabolic-cost data for a manuscript
- Engineered a signal-agnostic AR biofeedback platform (Python, Unity, HoloLens 2) pixel-streaming live via WebSockets to a head-mounted display for real-time gait feedback; sole developer, open-sourced
- Led first-author manuscript submitted to ICRA 2027 using OpenCV marker tracking to characterize AR display latency; found HoloLens 2 latency (120.2 ms) statistically equivalent to a wired monitor baseline (119.2 ms); designed and conducted a 2-subject pilot in which AR guidance reduced step-length asymmetry by 41% and 53%

Exoskeleton Research Intern

Locomotor Control Systems Laboratory, University of Michigan

Sept. 2026 – Present

Ann Arbor, MI

- Collecting and processing motion capture and EMG data from clinical trials evaluating a knee exoskeleton in patients with osteoarthritis and stroke
- Redesigning hip-belt geometry and dorsal motor-housing mounts to improve comfort and eliminate actuator-induced sag during extended wear

Cancer Biology Research Intern

University of Chicago Medicine

June 2025 – Aug. 2025

Chicago, IL

- Investigated β -tubulin isotype knockouts in bone-metastatic breast cancer cell lines using cell culture, adhesion assays, Western blotting, and confocal microscopy; presented findings to 100+ researchers at the University of Chicago Comprehensive Cancer Center Symposium

PROJECTS

Tibial Locking Compression Plate | SolidWorks, FEA, Simulink

Aug. 2026 – Sept. 2026

- Designed a parametric 10-hole Ti-6Al-4V locking compression plate with dimensions driven by global variables
- Validated to ASTM F382 via FEA in SolidWorks Simulation: 4-point bending (500 N) yielded FOS 1.73; torsion (10 N-m) yielded FOS 2.36; three-level mesh convergence confirmed stress converged within 3%
- Built a Simulink lumped-parameter healing model (plate and callus springs in parallel, logistic stiffness curve from literature) showing 46% plate stress reduction over 12 weeks

EMG-Controlled Prosthetic Gripper | SolidWorks, Arduino, Python

May 2025 – July 2026

- Designed and 3D-printed a 2-finger gripper in SolidWorks, integrating an MG996R servo, MyoWare 2.0 EMG sensor, and Arduino Uno with external power regulation to eliminate servo stutter under load
- Benchmarked two EMG classification pipelines across electrode-replacement sessions; achieved 99.4% accuracy across three forearm gestures using LDA with real-time end-to-end control

TECHNICAL SKILLS

Engineering: SolidWorks, SolidWorks Simulation (FEA), ASTM F382, 3D Printing, CNC Machining, Arduino, Simulink, TwinCAT, Vicon Motion Capture, EMG/ECG Signal Processing, LSL

Programming: Python, C++, MATLAB, Unity

Data & ML: OpenCV, WebSockets, NumPy, Pandas, SciPy, scikit-learn